11.2 Modelling the costing of projects

Topic Objective:

Understand the importance of modelling the costing of projects to achieve an optimum outcome. We will look at:

a. *budgets – undertake financial forecasts*
b. *planning for production – allocation of:*
    - *employees*
    - *materials*
    - *scale of production*
c. *selection of appropriate tools, machines and manufacturing processes.*

Research Task

Research your own examples of reasons for specific manufacturing methods being used for particular products.

You will find it easier to remember if you choose products related to your own interests.

For example:
Tennis rackets, football boots, gaming models, bicycles and fashion items.

For a high-performance component like a commercial aircraft wheel, the specific manufacturing method used is forging. This method is chosen over casting because the forging process involves deforming metal under great pressure, which creates a continuous grain flow that follows the contours of the wheel. This results in superior mechanical strength and fatigue resistance, which is critical for parts that must survive the immense impact and heat of landings. Additionally, forging ensures a higher level of structural integrity by eliminating internal voids or porosity, allowing the wheel to be both lighter and safer than one produced by traditional casting methods.

Now Test Yourself…

1. *Explain, using an appropriate product example, why a study of market trends would be an important element in a financial forecast.*

A study of market trends is a critical element of a financial forecast because it allows a company to predict future demand and adjust its economic projections accordingly. For example, in the electric vehicle (EV) industry, a manufacturer must analyze trends such as government subsidies and shifting consumer environmental concerns. If market trends indicate a rapid move away from internal combustion engines, a company's financial forecast

for a new petrol car might show declining revenue over time. By incorporating these trends, the company can make more accurate decisions regarding investment potential, retail pricing, and production levels, ultimately ensuring the long-term financial viability of their product line.

2. *Select and name a product that is made by die casting and give three reasons for selecting die casting rather than sand casting.*

Engine components, such as a cylinder head, are frequently made using die casting rather than sand casting for several technical and economic reasons. Firstly, die casting offers far superior dimensional accuracy and surface finish, often eliminating the need for expensive secondary machining processes. Secondly, it allows for the creation of complex, thin-walled sections that would be structurally impossible or too fragile to produce in a sand mould. Finally, while the initial tooling is expensive, die casting is much better suited for high-volume production, as the permanent metal dies can be reused thousands of times, making the cost-per-unit significantly lower than the labor-intensive sand casting process.

3. *Explain why car bodies for top selling models such as the Ford Fiesta are not made using mechanical fasteners and adhesives.*

Car bodies for top-selling, high-volume models like the Ford Fiesta are not assembled using mechanical fasteners and adhesives because these methods are too slow and costly for mass production. Instead, these vehicles are made from low carbon steel, which is exceptionally well-suited for robotic spot welding. Welding creates a permanent, high-strength bond very quickly and can be fully automated on an assembly line, ensuring consistency and low labor costs. While mechanical fasteners and specialized adhesives are often used for prestige cars made of aluminium (which is more difficult to weld), they do not offer the speed or cost-efficiency required to keep an economy model like the Fiesta competitively priced for the mass market.